

Question Title: Your First Visualization — Exploring Restaurant Tips

Problem Statement:

You are a junior data analyst at a restaurant chain. Your manager has asked you to create a few basic charts to explore customer tipping behavior. In this assignment, you will practice creating simple visualizations using Matplotlib and Seaborn for the first time.

Dataset:

Load the **Tips dataset** directly via Seaborn — no CSV download needed:

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
tips = sns.load_dataset("tips")
print(tips.head())
```

Relevant columns: total_bill, tip, day, size

Task 1 — Your First Matplotlib Chart

Using Matplotlib, create a **bar chart** that shows the **total tip amount collected on each day of the week**.

- x-axis: day, y-axis: total sum of tip
- Add a title, x-label, and y-label
- Set the figure size to (8, 5) using plt.figure(figsize=(8, 5))
- Call plt.show() to display the chart

Hint: Use tips.groupby("day")["tip"].sum() to get the total tips per day before plotting.

Task 2 — Seaborn Histogram

Using Seaborn, create a **histogram** of the tip column to see how tip amounts are distributed across all customers.

- Set bins=20
- Overlay a **KDE curve** on the histogram
- Add the title: "Distribution of Tip Amounts"
- Call plt.show() to display the chart

In a markdown cell below your chart, write **2–3 sentences** describing what the chart tells you — for example: where most tips fall, whether the distribution is skewed, and what the KDE line adds to the histogram.

Task 3 — Seaborn Scatter Plot

Using Seaborn, create a **scatter plot** to explore the relationship between the total bill and the tip amount.

- x-axis: total_bill, y-axis: tip
- Color the points by day using the hue parameter
- Add the title: "Total Bill vs Tip Amount"
- Call plt.show() to display the chart

In a markdown cell below, write **1–2 sentences** describing any pattern you notice — for example, does a higher bill generally lead to a higher tip?

Submission Guidelines:

1. Complete all tasks in a single .ipynb file with code cells, inline charts, and markdown explanation cells.
2. **GitHub:** Upload and submit a public repository link. **Google Colab:** Share with "Anyone with the link can view" access.